

Replicating Table 3 (Fivefold LATE) from Chernozhukov et al. (2018)

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GitHub Repository: <https://github.com/Seth-Ab/552-Replication>

1 The Original Paper

Chernozhukov et al. (2018) studies how Double/Debiased Machine Learning can be used to estimate causal parameters while allowing flexible machine-learning methods in the nuisance-function stage. The paper’s central concern is whether modern prediction tools can be combined with orthogonal score construction and cross-fitting in a way that preserves valid inference for treatment effects.

The application I replicate uses the 1991 SIPP 401(k) data. In that exercise, the paper estimates the effect of 401(k) participation on net financial assets while treating participation as endogenous and using 401(k) eligibility as an instrument. That setting makes Table 3 a natural illustration of the interactive instrumental-variables version of Double Machine Learning.

2 The Result of Interest

I replicate the fivefold cross-fitting row of Table 3 from Chernozhukov et al. (2018). More specifically, I target the row reporting the Local Average Treatment Effect of 401(k) participation on net financial assets across the paper’s set of nuisance learners. This is the most useful result to reproduce because it is the paper’s main empirical IV example and because it shows whether the qualitative conclusion is robust to different first-stage machine-learning choices.

3 Data and Methods

The raw data are stored in `input/sipp1991.dta`, the 1991 SIPP household sample used in the DoubleML 401(k) application. The variables used downstream are documented in `input/data_dictionary.md`. The key variables are net financial assets as the outcome, 401(k) participation as the treatment, 401(k) eligibility as the instrument, and a set of demographic and financial controls.

The replication pipeline is split into preprocessing and analysis. The script `code/preprocess.py` reads the raw Stata file from `input/`, keeps the variables used in the replication, drops missing values, and writes a deterministic analysis-ready file to `temp/clean_data.csv`. The script `code/analysis.py` then reads that cleaned file, estimates the IIVM specification with fivefold cross-fitting, writes the tracked output tables into `output/tables/`, and prints a console comparison between the published row and the replicated row.

4 Replication Results

Table 1: Table 3 fivefold LATE: paper targets versus simple Python replication

Method	Paper LATE	Paper [SE]	Simple LATE	Simple (SE)	Difference
Lasso	8944	[2259]	2389	(5190)	-6555
Random Forest	11764	[1788]	12639	(2189)	+875
Regression Tree	11459	[1717]	10682	(2149)	-777
Boosting	11133	[1661]	11462	(1624)	+329
Neural Network	11186	[1795]	11379	(2068)	+193
Ensemble	11173	[1641]	9710	(2644)	-1463
Best	11113	[1645]	11462	(1624)	+349

Notes: Paper targets are the fivefold LATE row from Table 3 of Chernozhukov et al. (2018). The simple replication uses one fivefold split from `simple.py`, not the paper’s 100-split median aggregation. The simple script’s Ensemble and Best rows are simplified constructions and should not be interpreted as exact replicas of the published definitions.

The generated table reports the published targets in Panel A and the replication results in Panel B. The main qualitative pattern carries over: the estimated LATE remains positive across learners, and the broad magnitudes are comparable to the original table. The replication therefore succeeds at reproducing the paper’s central empirical message even though it does not match every learner exactly.

The remaining discrepancies are plausible given the structure of the exercise. Learner-specific tuning choices can move nuisance predictions substantially, especially for penalized

and neural-network specifications. In addition, even when the estimand and cross-fitting structure are aligned, implementation details in repeated sample splitting and aggregation can affect the final row.

5 What I Learned

The easiest part of the replication was identifying the target result once I committed to reproducing only one row instead of the entire paper. The hardest part was making the machine-learning nuisance stage stable and reproducible across several learners while keeping the full pipeline automated from raw data through the final PDF.

The main lesson from the exercise is that reproducibility depends as much on workflow discipline as it does on the estimator itself. A clear directory structure, deterministic preprocessing, generated LaTeX fragments, and a real Makefile DAG make it much easier for another researcher to understand what was done and to rerun the analysis without guessing. If I were the original author, the most helpful additional detail would be more explicit documentation of the learner tuning and repeated-split aggregation choices used for the final table.

References

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.